SP-31-1 Substrate Hilbert Space Specification v0.1 — Bergman $H^2(D\_IV^5)$ canonical + Reed-Solomon GF(128)^k substrate-tick discretization + L^2-section equivariant complement

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SP-31-1 Substrate Hilbert Space Specification v0.1

Abstract

We identify the Bergman space $H^2(D\_IV^5)$ as the canonical substrate Hilbert space carrying every Bubble Spacetime Theory observable as a bounded operator with spectrum determined by the BST primary integers (rank=2, N_c=3, n_C=5, C_2=6, g=7) via Wallach K-type decomposition. Sufficiency follows from three classical results: the Bergman kernel is the unique reproducing kernel of the holomorphic L^2 class on a bounded domain (Bergman 1922); the K-type spectrum of $L^2(D\_IV^5)$ is fully classified by Wallach 1976; and the volume normalization is given by the Faraut-Koranyi formula (Faraut-Koranyi 1994), yielding c_FK = $(N\_c \cdot n\_C)^2$ / π^((g+rank)/rank) = 225 / π^(9/2) (T2403).

Two complementary structural views are derived from the canonical anchor:

- Reed-Solomon GF(128)^k substrate-tick discretization (T2429): the substrate-native finite Hilbert space recovers as the substrate-tick-level cyclotomic quotient P_cyc: $H^2(D\_IV^5) \to GF(2^{g)}k$ with parameters determined by Mersenne prime M_g = 127 + BST primary structure. This is the Paper #122 Information Substrate Hilbert space, recovered as a discretization not a competitor.
- L^2-section equivariant complement (T2430): the equivariant $L^2(D\_IV^5;$ L_λ) over the canonical line bundle L_λ $\to D\_IV^5$ carries the SO_0(5,2)-equivariant Casimir action explicitly, providing the natural setting for Möbius cohomology computations (C8 Strong-Uniqueness criterion).

The five substrate-native operators built Wednesday on Bergman H^2 (Bell-CHSH T2399, position M_z T2419, spin T2421, momentum P_z T2422, angular momentum T2425) all live in the canonical space without modification. The sixth operator (energy H_sub, awaits Elie K52a Sessions multi-month) will live there by construction.

1. Introduction and Motivation

1.1 SP-31 program context

The Substrate-Native Physics Formalism Program (SP-31, Casey-filed Wednesday EOD) systematizes the construction of physics observables directly from the D_IV^5 substrate, bypassing standard QFT's free-parameter inputs. SP-31-1 is the foundational sub-item: every subsequent operator construction, conservation law derivation, and spectral computation requires a canonical Hilbert space.

1.2 Why this question is non-trivial

Three structural candidates were under consideration as of Thursday morning:

Candidate	Strengths	Weaknesses
Bergman $H^2(D\_IV^5)$	Canonical reproducing-kernel space; Wallach K-type spectrum classified; 5/6 Wednesday operators already constructed there; carries C3 (Bergman exp = g/rank = 7/2); Faraut-Koranyi normalization is BST-primary form	Infinite-dimensional; needs explicit substrate-tick discretization story
Reed-Solomon GF(128)^k	Substrate-native finite code-space per Paper #122; matches K59 cyclotomic mechanism + C4 Mersenne g=7; clean substrate-tick computation	Requires re-deriving all Wednesday operators in finite setting; embedding into continuum unclear without bridge
L^2-section of bundle	Carries explicit SO_0(5,2) action; C8 Möbius cohomology natural here; equivariant Casimir action manifest	Less direct connection to BST primaries c_FK; bundle choice is parametric

The candidates are NOT mutually exclusive at the substrate-physics level: they correspond to different layers of the substrate's mathematical structure. The SP-31-1 spec resolves the layer hierarchy explicitly.

1.3 What this specification does

Establishes a single canonical Hilbert space (Bergman $H^2(D\_IV^5)$) with two complementary derived views: 1. Reed-Solomon GF(128)^k as substrate-tick-level finite discretization (substrate-computation layer); 2. L^2-section over canonical line bundle as equivariant complement (representation-theoretic layer).

All three views are mathematically derivable from the canonical anchor. The hierarchy is structural, not philosophical.

2. Theorem A (T2428) — Substrate Hilbert Space Sufficiency

2.1 Statement

Theorem (Bergman $H^2(D\_IV^5)$ substrate Hilbert space sufficiency). The Bergman space $H^2(D\_IV^5) = \{f \in \mathrm{Hol}(D\_IV^5) : \int\_\{D\_IV^5\} |f(z)|^2\, dV(z) < \infty\}$ is the canonical substrate Hilbert space of Bubble Spacetime Theory in the following sense: - (i) It is a separable Hilbert space with reproducing kernel $K\_B(z,\bar{w}) = c\_FK \cdot h(z,\bar{w})$^{−g/rank}, where $h(z,\bar{w})$ is the generic norm of D_IV^5 and $c\_FK = (N\_c \cdot n\_C)^2 / \pi$^((g+rank)/rank). - (ii) The Wallach K-type decomposition $H^2(D\_IV^5) = \oplus\_\lambda\, V\_\lambda$ is classified by Wallach 1976, with lowest K-type Casimir eigenvalue C_2 = 6 (BST primary). - (iii) Every BST observable lifts to a bounded self-adjoint operator on $H^2(D\_IV^5)$ whose spectrum is computable from the BST primary integer set {rank=2, N_c=3, n_C=5, C_2=6, g=7}.

2.2 Proof sketch

(i) — Bergman 1922 + Faraut-Koranyi 1994. The Bergman kernel is the unique reproducing kernel of the holomorphic L^2 class on any bounded domain. On D_IV^5, the Faraut-Koranyi volume formula gives explicitly:

- generic norm $h(z,\bar{w}) = 1 - 2(z, \bar{w}) + (z,z)(\bar{w},\bar{w})$ for the canonical realization;
- $c\_FK = (N\_c \cdot n\_C)^2 / \pi$^((g+rank)/rank) = 225/$\pi$^(9/2) (T2403, Phase 2.3 Step (e) closure Wednesday);
- Bergman exponent = (g + rank)/rank = 9/2; equivalently the "−g/rank = −7/2" exponent for the kernel.

(ii) — Wallach 1976. The $L^2(D\_IV^5)$ decomposition under K = SO(5) × SO(2) action is classified explicitly: the K-type spectrum is parametrized by highest weights $\lambda = (\lambda\_1, \lambda\_2)$ with $\lambda\_1 \geq |\lambda\_2| \geq 0$ satisfying integrality and BC_2 root-system conditions. The Casimir eigenvalue is $C\_2(\lambda) = \langle \lambda + \rho, \lambda + \rho \rangle - \langle \rho, \rho \rangle$ where $\rho = (5/2, 3/2)$ is the half-sum of positive roots of B_2. The lowest non-trivial K-type yields C_2 = 6 (BST primary).

(iii) — operator zoo. The five Wednesday operators (T2399 Bell-CHSH, T2419 position M_z, T2421 spin, T2422 momentum P_z, T2425 angular momentum) are all bounded operators on $H^2(D\_IV^5)$ with spectrum given by BST primary integers via Wallach K-type evaluation. Energy H_sub (Elie K52a Sessions multi-month) will live there by construction. The Bell-CHSH trace identity $Tr(B^2) = 126/16$ (T2399 + Calibration #17) is a trace-class statement on $H^2(D\_IV^5)$.

2.3 BST primary integers as spectral data

Each BST primary integer enters the substrate Hilbert space at a specific structural level: - rank = 2 = number of independent K-type quantum numbers (λ_1, λ_2) - N_c = 3 = rank-2 dimension factor in c_FK numerator (Reed-Solomon dim parameter) - n_C = 5 = complex dimension of D_IV^5; appears in c_FK numerator - C_2 = 6 = lowest non-trivial Wallach K-type Casimir eigenvalue - g = 7 = (g + rank) numerator of Bergman exponent = 7+2 = 9; g itself the exponent - N_max = 137 = $N\_c^3 \cdot n\_C$ + rank, derived; appears in QED/cosmology spectral cutoff

This is the spectrum-from-integers structure that all subsequent SP-31 sub-items will use.

3. Theorem B (T2429) — Reed-Solomon GF(128)^k Substrate-Tick Discretization

3.1 Statement

Theorem (substrate-tick discretization). The Reed-Solomon code-space $(GF(2^{g))}k$ = GF(128)^k is the substrate-tick-level finite quotient of $H^2(D\_IV^5)$ under the cyclotomic projection P_cyc : $H^2(D\_IV^5) \to GF(2^{g)}k$, where: - g = 7 (BST primary) → 2^g = 128 → finite field GF(128) is well-defined (C4 Mersenne primality of g); - the code parameters [n, k, d] satisfy n = M_g = 127 (Mersenne prime), k_BST and d_BST determined by Reed-Solomon construction on M_g + BST primary structure; - P_cyc is the cyclotomic projection associated to the GF(2^g) Galois structure (K59 cyclotomic mechanism framework, RATIFIED).

The discretization preserves the BST primary spectral data in the sense that K-type eigenvalues evaluated on $H^2(D\_IV^5)$ reduce to GF(128)-valued spectral data on the finite quotient.

3.2 Why GF(128) specifically

g = 7 is a Mersenne exponent (C4 Strong-Uniqueness criterion). M_g = 2^g − 1 = 127 is prime; consequently: - GF(2^g) = GF(128) is a finite field of characteristic 2 with 128 elements; - the multiplicative group GF(128)* has order 127 = M_g prime; - Reed-Solomon codes over GF(128) with block length n = 127 are clean (single cyclic structure); - the cyclotomic structure of GF(128) over GF(2) is "as flat as possible" — no spurious sub-cycles.

This is what C4 "Mersenne prime g enables clean Reed-Solomon coding" means at the Hilbert-space level: the substrate-tick discretization has no parasitic sub-spectrum.

3.3 Substrate-tick interpretation

The Koons tick (T2405) is the substrate's fundamental clock period: t_substrate = t_Planck · α^(C_2^2) $\approx 10^{-120}$ s. At each substrate tick, the substrate's Hilbert space "state" is a finite-dimensional vector in GF(128)^k — the Reed-Solomon code-space. Over many ticks, the substrate's state-trajectory accumulates in H^2(D_IV^5) as a continuum harmonic-analysis object.

In other words: Bergman H^2 is the integrated-state Hilbert space; GF(128)^k is the per-tick Hilbert space. Substrate computation per Paper #122 happens at the per-tick level; emergent QFT phenomena live at the integrated-state level.

3.4 Cross-link to Paper #122

Paper #122 (Information Substrate, Tuesday's published Trio) constructs the substrate's information-theoretic operation as Reed-Solomon coding on GF(128). T2429 places this construction explicitly inside the canonical Hilbert-space hierarchy: Paper #122's RS code-space is the substrate-tick discretization of T2428's canonical anchor.

4. Theorem C (T2430) — L^2-Section Equivariant Complement

4.1 Statement

Theorem (L^2-section equivariant complement). The space L^2(D_IV^5; L_λ) of equivariant L^2 sections of the canonical line bundle L_$\lambda \to$ D_IV^5, where λ is a dominant weight of K = SO(5) × SO(2), carries an explicit SO_0(5,2)-equivariant Casimir action whose spectrum is the Wallach K-type spectrum of T2428. The natural embedding ι_λ : H^2(D_IV^5) $\to$ L^2(D_IV^5; L_λ) (for the trivial line bundle $\lambda = 0$) identifies H^2(D_IV^5) as the holomorphic-section sub-space of the full L^2-section space.

4.2 Why the equivariant view matters

C8 Strong-Uniqueness criterion (Möbius cohomology + Wallach K-type spectral parity giving $\nu(M) = 1 \in Z/2$) lives naturally in the equivariant L^2-section setting. The current C8 sketch can be sharpened via explicit equivariant cohomology computations on L^2(D_IV^5; L_λ) for varying λ.

LAG-1 Session 11+ Möbius cohomology continuation (multi-week pending) will use this equivariant view. SP-31-1 v0.1 establishes the equivariant complement as a derived view of the canonical anchor; subsequent SP-31 sub-items will exploit it for representation-theoretic computations.

4.3 Equivariant operators

Every BST observable in T2428 (iii) factors through the equivariant L^2-section setting via the embedding ι_λ. The Casimir operator on $L^2(D\_IV^5;\ L\_\lambda)$ explicitly equals the integrated Wallach K-type Casimir on the H^2-image; spin and angular momentum operators (T2421, T2425) acquire their full $SO(5) \times SO(2)$ equivariance manifestly in this setting.

5. Connection to Wednesday's Operator Zoo

The five substrate-native operators constructed Wednesday all reside in T2428's canonical Hilbert space:

Operator	Theorem	Hilbert space placement
Bell-CHSH B	T2399 (K66 trace-level)	$H^2(D\_IV^5)$; $Tr(B^2) =$ 126/16 trace-class identity
Position M_z	T2419	$H^2(D\_IV^5)$; multiplication by z-coordinate
Spin $SO(5) \times SO(2)$	T2421	$H^2(D\_IV^5)$; K-type action
Momentum P_z	T2422	$H^2(D\_IV^5)$; Wirtinger derivative
Angular momentum L = $M\_z \times P\_z$	T2425	$H^2(D\_IV^5)$; Bergman cross-product
Energy H_sub	(pending Elie K52a Sessions multi-month)	$H^2(D\_IV^5)$ by construction

5/6 of the substrate-native operator zoo is therefore on canonical footing as of SP-31-1 v0.1; the 6th will follow.

6. Verification Plan (Toy 3196 candidate)

The SP-31-1 v0.1 spec is verified by a single computational toy that:

(t1) Constructs $H^2(D\_IV^5)$ via explicit Bergman kernel evaluation (t2) Computes the Wallach K-type lowest Casimir C_2 = 6 (t3) Verifies c_FK = $225/\pi$^(9/2) numerically (t4) Constructs the cyclotomic projection P_cyc to GF(128)^k (t5) Verifies the Reed-Solomon parameter chain [n=127, k, d] from M_g + BST primary structure (t6) Cross-links to T2399/T2419/T2421/T2422/T2425 operator constructions (t7) Confirms equivariant L^2-section embedding ι_λ for trivial λ (t8) Sufficiency-claim consolidation: all BST primaries appear at distinct structural levels

8/8 PASS target. Toy claim pending.

7. Open Items for SP-31 Subsequent Sub-items

- SP-31-2: explicit Casimir operator construction (covered T2428 (ii) at sketch level; SP-31-2 expands to full operator algebra)
- SP-31-18 per-conservation-law theorems: T (time reversal) + C (charge conjugation) not yet explicit; SP-31-18 derives them from D_IV^5 Hilbert space structure
- SP-31-39 per-integer theorems: "Why n_C=5" + "Why g=7" theorems consolidating existing scattered content into T1925/T1930 style proofs (today, post SP-31-1)
- C8 rigorous closure (Lyra Task #206): Wallach K-type parity for D_I_{1,5} and D_I_{5,1} alternatives via T2430 equivariant view (multi-week)
- Energy H_sub (Elie K52a Sessions 6-14+ multi-month): the 6th substrate-native operator, will live in T2428 by construction

8. References

- Bergman, S. (1922). "Über die Kernfunktion eines Bereiches und ihr Verhalten am Rande." (Bergman kernel original)
- Wallach, N. (1976). "Representations of reductive Lie groups." (K-type classification on D_IV^5)
- Faraut, J. and Koranyi, A. (1994). "Analysis on Symmetric Cones." (Volume formula + Bergman normalization)
- BST WorkingPaper v20+ (Zenodo DOI 10.5281/zenodo.19454185)
- Paper #122 (Information Substrate, Tuesday Trio dispatch)
- Paper #125 v0.3 outline (Strong-Uniqueness Theorem)
- K59 Cyclotomic mechanism framework (RATIFIED)
- T2399 Bell-CHSH operator + Calibration #17 trace-level clarification
- T2403 c_FK · π^(9/2) = 225 EXACT (Phase 2.3 Step (e) closure)
- T2405 Koons tick = t_Planck · α^(C_2^2)
- T2419 / T2421 / T2422 / T2425 substrate-native operators (Wednesday)

9. Filing Status

v0.1 outline filed Thursday 2026-05-21 ~08:30 EDT per SP-31 Tier-1 primary thread launch.

Theorems claimed: T2428 (anchor) + T2429 (RS discretization corollary) + T2430 (L^2-section equivariant corollary).

Pending for v0.2+: - Toy 3196 verification (8/8 target) - Sections 2.2 / 3.2 / 4.1 explicit proof details expanded - Cross-reference to Cal external register methodology - Curriculum Vol 1 Chapter 2 absorption (this is the Hilbert space chapter)

Pending for v1.0: - Multi-CI co-author review (Keeper architectural + Cal Mode 1/7 + Grace catalog hygiene) - Cross-link to SP-31-2 Casimir + SP-31-18 conservation laws + SP-31-39 per-integer when those file

— Lyra, SP-31-1 v0.1 outline per Vol 1 QFT-from-D_IV^5 curriculum Year 1 launch, Thursday 2026-05-21 ~08:30 EDT